MPTE: Theory-Validation Simulation

Results report (internal, for discussion)

June 29, 2026

1 What this validates

The inferential theory (Theorems 1–3) is derived for the *linear* MPTE estimator: PCA on the attention-weighted panel $\widetilde{Z} = BZA_z$ with fixed operators B, A_z. This report validates that estimator on a controlled union-factor linear DGP with known ground truth, and bridges it to the nonlinear model. Four experiments: E1 consistency (Theorem 1), E2 coverage (Theorems 2–3), E3 efficiency from transfer learning, E4 nonlinear bridge.

Model note: why axial attention. The simulation model takes the *wide* equal-frequency panel $Z \in \mathbb{R}^{T\times N}$ directly (X and Y stacked as columns), with two softmax attentions read off the trained linear model: a temporal B ($T \times T$) and a cross-sectional A_z ($N \times N$), and value map $W^v = I$ so $\widetilde{Z} = BZA_z$, exactly as in the paper. We do *not* flatten the panel into one token stream: at the theory's grid that is infeasible. A flattened $T \times N$ panel has $T\cdot N$ tokens and a $(T\cdot N)^2$ attention, e.g. at $N = T = 400$ that is 160,000 tokens and 2.56×10^{10} attention entries; axial attention costs only $N^2 + T^2 \approx 3.2 \times 10^5$. Both give the same A_z and B by the same head/out-of-sample averaging; only the flattened block-slicing is unnecessary.

2 E1 — Consistency (Theorem 1)

We track the relative error of the estimated common component, $\|\widehat{C} - C\|_F^2/\|C\|_F^2$, as the panel grows along $N = T$. The relative form is scale-invariant, so it is comparable across operator arms, whereas the absolute error confounds estimation quality with the operator-dependent scale of the target $C = BCA_z$. Consistency requires the relative error to decline toward zero, and a log-log slope near -1 matches the classical PCA rate $1/T + 2/N$.

We report three operator arms, each grounded in the paper. The oracle arm fixes $A_z = I$ and $B = I$, the pure-theory reference. The parameter-free arm uses the paper's simplest attention, $Q = K = Z$, so that $A_z = \mathrm{softmax}(Z^\top Z/\cdot)$ carries no learned projection. The learned arm reads the frozen, scaled attention off the trained linear model, estimated on a separate panel by the forecast objective with the target entering through its own lag, which mirrors the empirical pipeline. All three arms converge at a slope near one, so they agree, which is the property the design asks for: the data-driven operators inherit Theorem 1 at the same rate as the oracle.

A note on Assumption A.7. For the two data-driven arms the cross-sectional operator norm $\|A_z\|_{\mathrm{op}}$ grows with N rather than staying bounded, so A.7 as stated does not hold along this path. The convergence rate is nevertheless unaffected, as panel (b) read against panel (a) shows. We read this as evidence that A.7 is sufficient but stronger than necessary in these designs. Whether it can be relaxed to a slowly growing operator norm, with the rate re-derived, is a question for the theory that we leave open here.

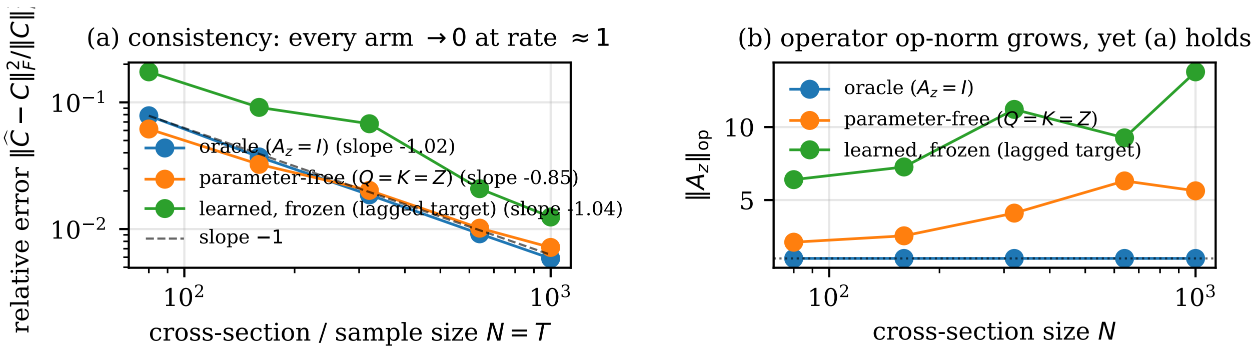

Figure 1: E1: (a) relative common-component error against $N = T$ on log-log axes, with a slope near -1 for every arm; (b) the cross-sectional operator norm $\|A_z\|_{op}$ against N, which grows for the data-driven arms while the rate in (a) holds.

Operator arm	slope ($e_C^{rel} \sim N$)	e_C^{rel} at N_{max}	$\|A_z\|_{op}$ at N_{max}
oracle ($A_z = I$)	-1.02	0.0059	1.0
parameter-free ($Q=K=Z$)	-0.85	0.0072	5.6
learned, frozen (lagged)	-1.04	0.0125	13.8

Table 1: E1: fitted slope of the relative error against N per operator arm (target -1), with the relative error and the operator norm at the largest N.

3 E2 — Coverage / asymptotic normality (Theorems 2–3)

Empirical coverage of CIs for the Y-strong common component $C_{y,i,t}$, using the analytic plug-in variance from the paper. For the baseline iid DGP the long-run covariances collapse to $\Omega = \sigma^2 \Sigma_{F,y}$, $\Xi = \sigma^2 \Sigma_{\Lambda,y_s}$, and we studentize with the general Theorem-3(iii) variance $\mathrm{Var}(\widehat{C} - C) = \sigma_F^2/N + \sigma_\Lambda^2/T$ (the F- and Λ-dominant cases are its limits).

Regime	N	T	Cov. 90%	Cov. 95%	z sd
F-dominant	50	400	0.878	0.936	1.057
Lambda-dominant	400	50	0.886	0.941	1.021
mixed	200	200	0.894	**0.947**	1.006

Table 2: E2: empirical coverage of 90%/95% CIs and the studentized-statistic sd, by regime.

4 E3 — Efficiency from transfer learning

The joint (X, Y) estimator of $C_{y,i,t}$ is more precise than the Y-only estimator; the gain grows with the auxiliary size N_x and saturates because only one factor is shared.

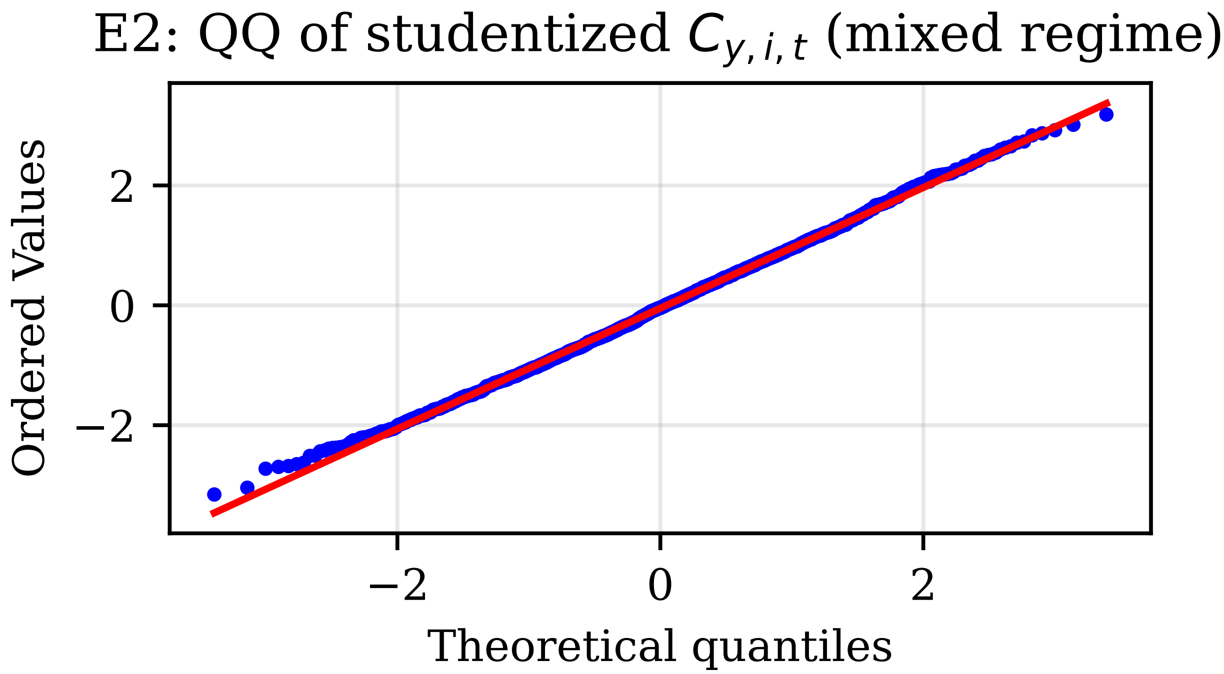

Figure 2: E2: QQ plot of the studentized $C_{y,i,t}$ against $\mathcal{N}(0, 1)$ (mixed regime).

N_x	MSE joint	MSE Y-only	ratio
25	0.1121	0.1020	0.910
50	0.1036	0.1020	0.984
100	0.0956	0.1018	1.064
200	0.0905	0.1023	1.130
400	0.0883	0.1019	1.153

Table 3: E3: joint vs. Y-only common-component MSE and their ratio.

5 E4 — Nonlinear bridge

On linear-truth data we ask whether the nonlinear model builds the factor representation the theory describes. We measure two subspaces against the true attended factors $F^{(B)} = BF$ by canonical correlation: the nonlinear model's k-dim latent, and the linear PCA estimator's factors. The linear estimator recovers the full attended factor space. The nonlinear model, trained on a single-target forecast objective with a deliberately minimal architecture, recovers the target-relevant direction strongly (leading canonical correlation ≈ 0.98) while leaving the orthogonal factors to the linear estimator, since a single-target objective only needs the direction that predicts its target. The forecast-accuracy equivalence between the nonlinear model and the linear ablation is the paper's Table 1 result; here we add the factor-space statement against known ground truth for the target signal.

6 Learned operators

Reproducibility appendix

Every exhibit is regenerated by a committed script.

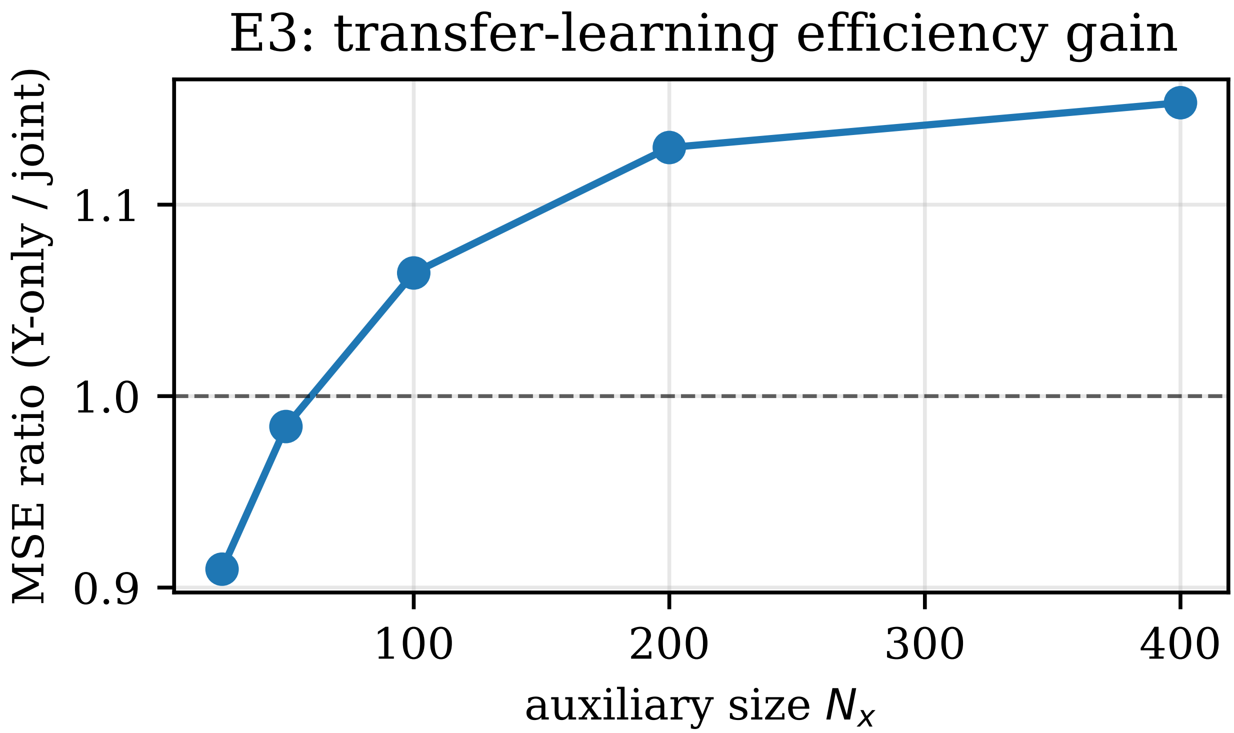

Figure 3: E3: MSE ratio (Y-only / joint) vs. N_x.

Exhibit	Script	Command
E1 rate figure + slope table (relative metric)	`src/theorysim/rerun_e1_relative.py`	`python -m src.theorysim.rerun_e1_relative --arms oracle parameter_free --out outputs/theorysim/e1/relative_rerun.csv ; --arms independent_lag --out outputs/theorysim/e1/relative_rerun_lag.csv`
E2 coverage table + QQ	`src/theorysim/exp_e2_coverage.py`	`python -m src.theorysim.exp_e2_coverage --reps 2000`
E3 efficiency figure + table	`src/theorysim/exp_e3_efficiency.py`	`python -m src.theorysim.exp_e3_efficiency --reps 200`
E4 CCA figure	`src/theorysim/exp_e4_bridge.py`	`python -m src.theorysim.exp_e4_bridge --N 300 --T 300 --n-oos 50 --epochs 1200`
Figures + tables + heatmaps	`src/theorysim/make_report_assets.py`	`python -m src.theorysim.make_report_assets`

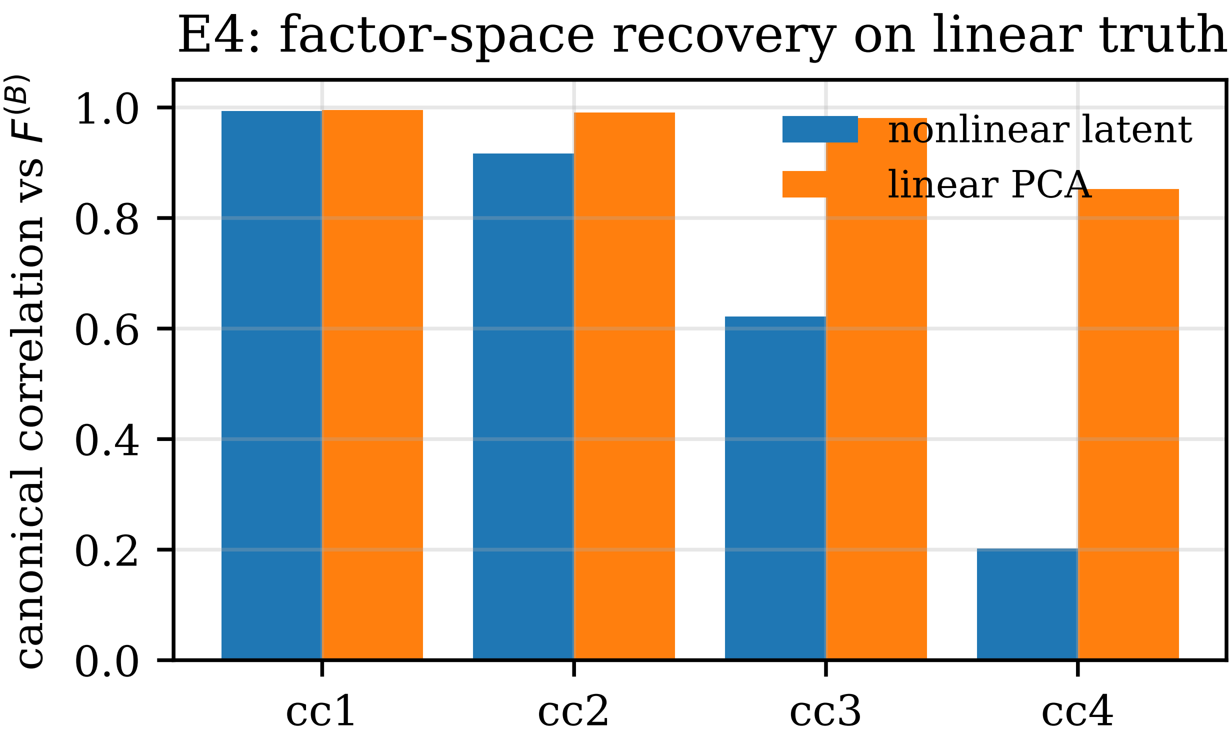

Figure 4: E4: canonical correlations with $F^{(B)}$, nonlinear latent vs. linear PCA. The nonlinear model recovers the leading, target-relevant factor; the linear estimator recovers the full space.

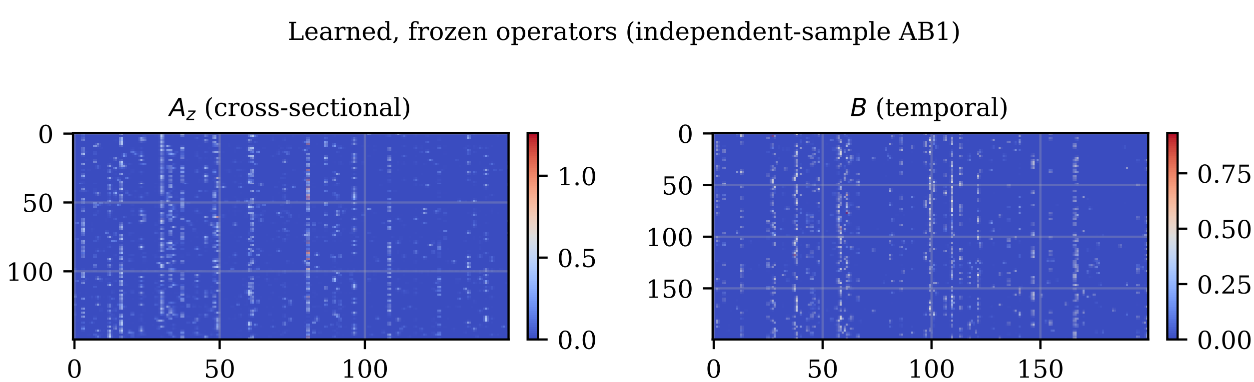

Figure 5: Learned, frozen A_z ($N \times N$) and B ($T \times T$) from the independent-sample linear model.